

Problem 2

Problem 1 State the form of the limit. Determine whether the form is determinate or indeterminate. Evaluate each limit.

(a) $\lim_{x \rightarrow \infty} (\ln(1 + e^{-x}))^x$

Explanation. Since $\lim_{x \rightarrow \infty} (1 + e^{-x}) = 1 + 0 = 1$ and $\ln(1) = 0$, this limit is of the form 0^∞ . This is a determinate form. $\lim_{x \rightarrow \infty} (\ln(1 + e^{-x}))^x = 0$.

(b) $\lim_{x \rightarrow \infty} \left(\frac{1}{x} + 1 \right)^{\frac{1}{x}}$

Explanation. This limit is of the form 1^0 , which is a determinate form.

Thus, $\lim_{x \rightarrow \infty} \left(\frac{1}{x} + 1 \right)^{\frac{1}{x}} = 1$

(c) $\lim_{x \rightarrow \infty} \left(\frac{\arctan(x)}{x} \right)$

Explanation. Since $\lim_{x \rightarrow \infty} \arctan(x) = \frac{\pi}{2}$, this limit is of the form $\frac{\#}{\infty}$, which is a determinate form.

Thus, $\lim_{x \rightarrow \infty} \left(\frac{\arctan(x)}{x} \right) = 0$

(d) $\lim_{x \rightarrow \infty} (x - \ln(x))$

Explanation. This limit is of the form $\infty - \infty$, which is an indeterminate form. We can rewrite this as:

$$\lim_{x \rightarrow \infty} (x - \ln(x)) = \lim_{x \rightarrow \infty} \left(x \left(1 - \frac{\ln(x)}{x} \right) \right)$$

We can see that $\lim_{x \rightarrow \infty} \left(\frac{\ln(x)}{x} \right)$ is of the form $\frac{\infty}{\infty}$, so we can use L'Hopital's Rule.

$$\stackrel{L.R.}{\implies} \lim_{x \rightarrow \infty} \left(\frac{1/x}{1} \right) = \lim_{x \rightarrow \infty} \left(\frac{1}{x} \right) = 0$$

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Now we have:

$$\lim_{x \rightarrow \infty} \left(x \left(1 - \frac{\ln(x)}{x} \right) \right)$$

This limit has the form $\infty \cdot 1$. This is a determinate form, and, therefore,

$$\lim_{x \rightarrow \infty} (x - \ln(x)) = \infty$$

(e) $\lim_{x \rightarrow \infty} \left(x \ln \left(\frac{1}{x} \right) \right)$

Explanation. As x approaches ∞ , $\frac{1}{x}$ approaches 0 from the right. So

$$\lim_{x \rightarrow \infty} \ln \left(\frac{1}{x} \right) = -\infty$$

Therefore, the limit in question is of the form $\infty \cdot -\infty$, which is a determinate form.

Thus,

$$\lim_{x \rightarrow \infty} \left(x \ln \left(\frac{1}{x} \right) \right) = -\infty$$

(f) $\lim_{x \rightarrow 0^+} (\sin(x) \cot(x))$

Explanation. Since $\lim_{x \rightarrow 0^+} \cot(x) = \infty$, this limit is of the form $0 \cdot \infty$. This is an indeterminate form.

Note: $\cot(x) = \frac{\cos(x)}{\sin(x)}$. So

$$\lim_{x \rightarrow 0^+} (\sin(x) \cot(x)) = \lim_{x \rightarrow 0^+} \cos(x) = 1.$$